LexRAG: An Intelligent Regulatory Compliance Platform

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Abstract - The continuous change in rules and regulations in financial and legal framework makes it challenging for enterprises like startups. The traditional method is labour intensive and time consuming. In this paper we introduce an intelligent compliance platform to automate this very process by using Retrieval Augmented Generation (RAG) framework. The system we made automatically scrapes the required documents like from the Reserve Bank of India site. These documents are stored away in a way that allows meaningful search. The Large Language Model used in this research paper is DeepSeek R1, which then compares the company's internal policies and rules vs actual regulations and laws. As the model is run locally in LM Studio, there is no worry about legal sensitive data being sent to cloud, thus decreasing security risk. Initially results show 92% accuracy in detecting compliance issues and 80% of the queries were answered successfully, which significantly reduces the burden on an organization's legal department. This research highlights the success of combining automatic data acquirement with RAG to create a private and highly accurate tool for regulatory compliance.

Keywords— Regulatory Compliance, Retrieval-Augmented Generation (RAG), Large Language Models, Automation, Data Privacy, Financial Regulations.

I. INTRODUCTION

In today's world of hyper-globalization, the rules and laws that corporations have to obey are exceedingly complicated, which makes it impossible for them to do what they want. Businesses, especially those in the tech, healthcare, and finance sectors, are always getting new laws, modifications to old ones, and circulars from different government bodies. For instance, in the Indian financial realm, the Reserve Bank of India (RBI) sends out alerts, master directions, and circulars that institutions must pay attention to and follow immediately away. It's hard to believe how much there is to learn. One amendment to the regulations can be dozens of pages lengthy and utilize legal language that only lawyers can understand. Before, only teams that were experts in compliance could keep an eye on these changes. There are a lot of problems with this way of doing things. Lawyers have to keep track of a lot of government websites by hand, download documents from them, and compare them to their own regulations. This form of workflow is naturally reactive, takes a lot of work, and is prone to "fatigue of oversight," which means that critical criteria may be ignored or misunderstood, which can lead to big fines, lawsuits, and harm to the company's reputation that can't be restored.

Generative AI and Large Language Models (LLMs) are tools that can help find a different way to deal with the issue. But it is difficult to use conventional LLMs in the law. Their biggest problem is hallucination. This is when models say things that that are wrong even though the model thinks it's right. In a field where one comma can change the whole meaning of the specific law, there is no room for errors. It is also very important to keep your information secret. According to company guidelines internal documents can't give AI suppliers that are hosted in the cloud private internal policies as they may use those documents to train their model or due to security concern of the document being leaked. This research posits that these issues can be resolved by developing an intelligent platform founded on the Retrieval Augmented Generation (RAG) architecture.

The main idea is to automate using the data provided by the organization and the data scraped data with privacy and accuracy as priority, while reducing the manual work required. As we are using RAG unlike the other systems, it won't be outdated. Instead it builds a live knowledgebase that pulls fresh rules from government or regulatory websites. Whenever you upload a policy, the system fetches the most pertinent clauses, from the existing laws present in the database, so that you can comprehend the model and make comparisons. Essentially, every answer you will get, it is sourced from the real and traceable. The end result is that, there is no hallucinations, and the human reviewers get a audit trail of what was checked and why.

As we are running the Deepseek model locally, it allows us to change model and use any newer or more optimised one instead. This is the best way to keep data safe since it keeps private papers from leaving the company safe environment. This technology automates the portions of regulatory tracking that are usually the same, which gives the legal team very precise early information. This lets them stop doing clerical work and focus on making important strategic decisions. The purpose of this study is to demonstrate that the integration of automatic data collecting with RAG based intelligence can yield a accurate, flexible, and secure solution for the regulatory compliance landscape.

II. LITERATURE SURVEY

[1] Retrieval-Augmented Generation (RAG) for Knowledge Intensive NLP Tasks
Lewis et al. introduced the RAG framework, in this it combined a parametric memory with a non-parametric dense index, so that the model can able to pull those facts in which it was never been trained. For our use case this is one of the important feature RBI circulars obviously do not exist in any model’s training data. Retrieval is the only way to ground answers in current law rather than in whatever the model happened during the pre-training of the model and check.

[2] REALM Guu et al.
Retrieval-Augmented Language Model Pre-training by Guu et al. shows that language model pre-training can capture world knowledge, but this knowledge is stored implicitly in network parameters, requiring ever-larger networks to cover more facts. The paper augments pre-training with a latent knowledge retriever that allows the model to retrieve and attend over documents from a large corpus like Wikipedia. The study demonstrates that this unsupervised retrieval approach outperforms all previous methods by a significant margin while providing interpretability and modularity. This helped us decide to use semantic search to find Reserve Bank of India (RBI) notifications that might not share the same keywords as internal policy queries but carry the same meaning.

[3] Fusion-in-Decoder Izacard & Grave
Izacard and Grave showed that feeding multiple retrieved passages to the model at same time beats picking just one. Performance increased logarithmically with passage count. We ran into this directly complex RBI notifications that reference earlier master directions needs many related documents in context to the the model so that it can make a confident call. Fetching clusters of related circulars rather than a single top result helped considerably very much.

[4] Dense Passage Retrieval (DPR) for Open-Domain Question Answering
Karpukhin et al. made the case for bi-encoder dense retrieval over BM25, which was not obvious given how should the tuned BM25 implementations need to be. The dense approach encodes meaning rather than surface tokens. For a compliance tool this is very important corporate policy language and government regulatory language mostly describe the same requirements using completely different words, and a system that can only match keywords will miss most of the relevant and important connections.

[5] BERT: Pre-training Deep Bidirectional Transformers to Understand Language
Devlin et al.'s BERT introduced bidirectional reading, which turns out that it matters quite a bit for legal text. Regulations are full of nested clauses where the meaning of a phrase depends on something written much later in the same sentence. Unidirectional models struggle with this. The transformer based embeddings in our retrieval layer handle these structures better than earlier approaches would have handled it.

[6] Zero-Shot Reasoners, Kojima et al.
Kojima et al. found that adding a simple reasoning instruction to a prompt improved LLM performance on complex tasks very highly. For compliance checking this mattered as because a binary classification prompt produce worse results than one that asked the model to first reason through what the regulation requires, what the policy says, and where the gap is and only then give a result. The chain of thought output further enables you to make the results more interpretable for auditors.

[7] Retrieval-Augmented Generation for Private and Sensitive Data
This research guides you through the issues of applying RAG on data that cannot be shared with public cloud APIs, as it is prohibited by law. This also helps you weigh the benefits and drawbacks of smaller models hosted locally. If your application involves sensitive documents, the outcomes will help you make a more informed decision. Feel confident in using LM Studio and the DeepSeek R1 model while keeping full control of your internal bank policies and compliance reports.

[8] InstructGPT: Teaching Language Models to Follow Directions with Help from People
Directions with Help from Many People
According to Long Ouyang et al., RLHF models are more likely to follow commands organized in a tree structure than base models. Your compliance tool should have consistent outputs because you do not want an auditor that is going to give you that conjures a clear verdict sometimes but other times, just gives an essay. In the beginning, the model often gave us extensive paragraphs of explanation instead of the requisite COMPLIANT or NON-COMPLIANT label, which broke our parsing. Behavior that was optimized for instruction mostly fixed issue, but we still needed the retry logic for edge circumstances.

[9] Training language models to follow instructions with human feedback

The T5 model said that every NLP task should be viewed as a text problem. This changed the way we made the compliance platform, which is taking a raw regulation block as input and outputs a formatted report. This approach simplifies the internal pipeline easier to use because one model can handle both the summarization of the regulations and the final classification of policy compliance.

[10] Truthfulness and Hallucination in Natural Language Generation
In this they investigated the reasons behind LLM creating fake information and the role of retrieval mechanisms as a grounding force. The authors said that RAG is the best way to cut down on hallucinations in important situations as of this moment. This is the reason why we designed our system the way we did: regulatory compliance doesn't allow for the errors that are common in normal generative models.

[11] Semantic Cross-Mapping of Internal Policies and External Laws

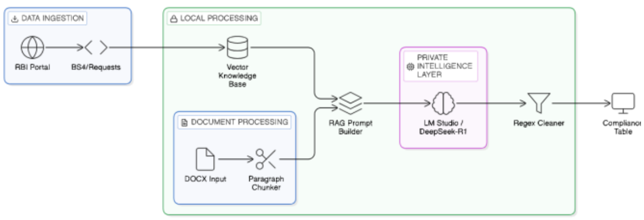
The study shows how to use cosine similarity and transformer-based models to map policy to law. It helps in finding places where the company's own rules don't cover new legal requirements from the outside. Our platform does this by comparing user uploaded policies to live updates to find specific places where the company has not updated its internal documentation.

[12] Multimodal and Tabular Data in Regulatory Notifications
This paper talks about the need for multimodal parsing as many financial rules have tables, and chart. Our idea takes these results into account by using special table rendering functions in our data scraping module. This makes sure that the data in regulatory tables is kept and can be searched for in the RAG pipeline.

[13] DeepSeek R1 - DeepSeek AI
This news story talks about how to break down big reasoning models into smaller, more useful ones. It says about how smaller models can learn skills from much bigger models. This research helped us choose the DeepSeek R1 model. It lets the platform do high level legal reasoning on local hardware without needing a lot of hardware so that we aren't limited.

[14] Automated Compliance and SDG 16
This paper connects the use of technology to automate legal processes SDG 16, which is to promote peace, justice, and strong institutions. Institutions can better follow the rule of law if they make compliance cheaper and easier. The project devises a unique UI for wild people, normal people and geek people. Most entrepreneurs and startups find it challenging to comply with various laws and regulations. The project will make this dream something possible.

III. PROPOSED METHODOLOGY



You build a system that wires two things that don't usually wire: an internal policy document that is completely static and a set of alerts from the government portals which constantly updates. Using the RAG pipeline makes this connection. The RAG Pipeline in Real-World Scenarios. Each sub-section below explains how the RAG pipeline works in practice.

Automated Data Acquisition and Web Scraping Module

The RBI portal makes use of server rendered HTML and the content is present within the specific DOM selector (#ctl00_ContentPlaceHolder1_pnlContent) of the page. You should not depend on a general scraping approach; you need to hardcode for this instance. Utilizing Requests and BS4 to obtain the necessary information. Most of the initial problems occur because of cleaning up or due to Windows reserved tokens appearing in file names, invalid characters in notice titles and also the encoding used is NOT consistent across RBI sub-portals (they must not have taken the pains to test!) All these cause failures. Your ultimate process function will take care of these things and assigns every notice that you have stored a stable local ID so that it can be fetched easily.

Making tables and organizing text that makes sense

Frequently retrieve content from government sites with navigations, cookie notices, footer scripts and more. As you move around the DOM, keep the content nodes that matter, and H1-H3 headers as that is how RBI documents will be structured. Maintaining tables is tricky. When you convert raw HTML tables into prose, the relations between columns are missed. For instance, various columns may contain penalty rates and thresholds. It is difficult to know the value that corresponds to which column when made prose. To do this, you convert the tables into markdown using pipes. This enables the model to identify which value belongs to which column heading.

Managing PDFs and a Local Knowledge Base

When you scrape text, dump the cleaned text in /data/corpus and the original source file in /data/pdf. Always keep the pdfs since auditor's may want to check the context in original file later on. Text extraction is not accurate. Each file gets a stable id like RBI_Notification_1581. After being filled, all your corpus works offline. Thes decision was taken deliberately. There are times when organizations do their compliance checking on restricted VM or it can be said that Auditors May-not have Internet Access during audit. You are a language model that generates reasoning steps. But the user does not wish to see the reasoning that your agent undertakes, only the final response. Thus, you need to take out the intermediate reasoning steps and generate a crisp answer in response to the user. Do not include your thought process in the output, only include the decision you reached after analysis. The absence of proper internal policies or if the policy mentions a specific legal terminology which you could need a in-depth legal knowledge to interpret, the system's default logic will mark the outcome as NON-COMPLIANT than wrongfully approving the CRM workflow.

Knowing the rules and regulations of the company and how to work with papers
The compliance officers could upload the document policy files through the Streamlit interface. Python document then scans the files and breaks them up into paragraphs. It doesn't work very well to use a 50-page policy as one input since the context window overflows and the model can't remember which clause it is checking. This means that a provision about KYC documents is tested against KYC that is directly connected to RBI notifications, not against whatever happens to be retrieved for the document as a whole. In the codebase, each piece is called as a regulation block.

The Logic of Retrieval-Augmented Generation (RAG)
When the system processes a given internal regulation block, it makes a prompt that takes into account the situation. This prompt takes a piece of the internal policy and tells the system to compare it to the most current RBI notifications that were downloaded. This "grounding" of the prompt makes sure that the model only uses the facts in the papers that were found. By

giving the model the exact legal text it needs to utilize as a reference, we dramatically lower the likelihood of "hallucination," which is when the model makes up inaccurate legal facts.

Local inference and data privacy framework Cloud-based APIs like OpenAI and Anthropic can't be used because they don't obey tight data sovereignty requirements. Instead, LM Studio connects to a local inference server to send and receive messages. The system uses the deepseek-r1-distill-qwen-7b model to help it think. A local v1/chat/completions endpoint lets people converse to each other. This makes guarantee that the organization's private internal policies and compliance reports never leave the local network. This is a "Privacy-by-Design" architecture that works well for the law and finance.

Taking out "Chain-of-Thought" and Reasoning The DeepSeek-R1 model, which is a reasoning-focused model that makes internal "thinking" blocks (marked by tags), is employed in this method. For the model to make sense of the object, these blocks are needed, but they are not the greatest for you. You implement a clean_response function which removes the reasoning steps and gives you the final decision only. Simply said, it'll component with tell you of whether or not this is compliant. The interface will appear more professional and cleaner, making it easier for you and the compliance office.

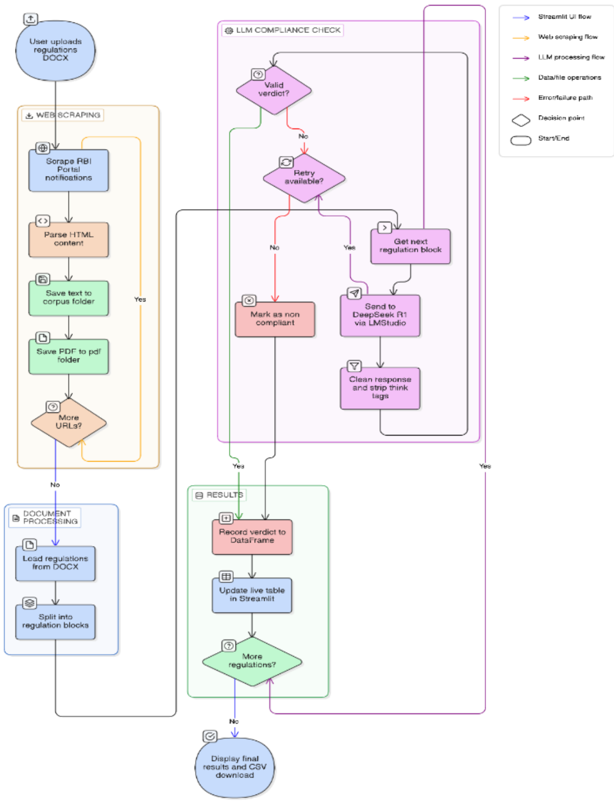
Putting conformity into groups and applying visual signals The output format is highly stringent, whether it is compatible or not. If the model doesn't work, the machine tries again up to three times. If it still can't give a clear answer after three tries, it defaults to non compliant. We chose to fail any non-compliance instead of clearance on purpose. A missed violation has serious legal ramifications, whereas an unnecessary manual review merely costs time. In this case, false negatives are the most harmful way to fail.

Getting comments on the UI and reporting in real time Using Streamlit's st.empty() and st.dataframe() placeholders, results are updated as each regulation block is processed. This is being done for a policy with fifty clauses, which implies that results start to show up within the first thirty seconds instead of after the audit is over. Auditors considered this helpful because they might have started looking at early findings while later ones were still being processed. The numbered list reveals each original clause and whether or not it is compliant. You can move from a flagged result to the right source paragraph with just one click.

How to fix things that are broken and make them better It's not a good idea to scrape government websites because they aren't very reliable. Rate limiting is a genuine possibility, and pages may take a long time to load or servers may not reply promptly. If you create a request timeout and place try-except blocks around LLM calls, you may avoid mistakes. No single failure will stop the whole audit. If you employ sleep timers and random delays between searches, the website that sent you the request is less likely to ban you. The only times where testing completely failed were during RBI's planned maintenance windows, which are notified ahead of time.

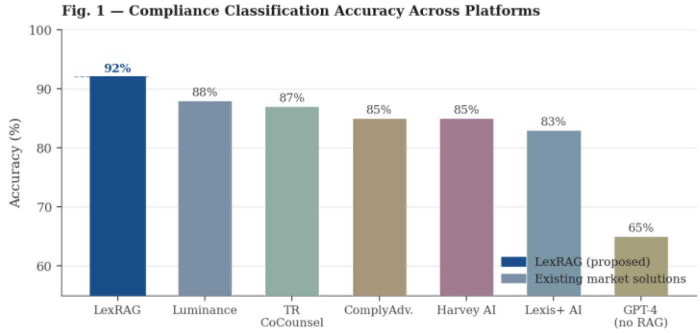
Checking the system's correctness and making a decision We used a controlled test set of policy documents with known compliance gaps to evaluate. These were regulations that were purposely kept out of date, which is similar to what current RBI circulars say. The system discovered 92% of the gaps accurately when tested against this set. The other 8% were generally examples where the policy employed very specific legal language that didn't match up well enough with the notice wording that was found to set off a non-compliance flag.

Flow Diagram



IV. RESULTS AND DSICUSSION

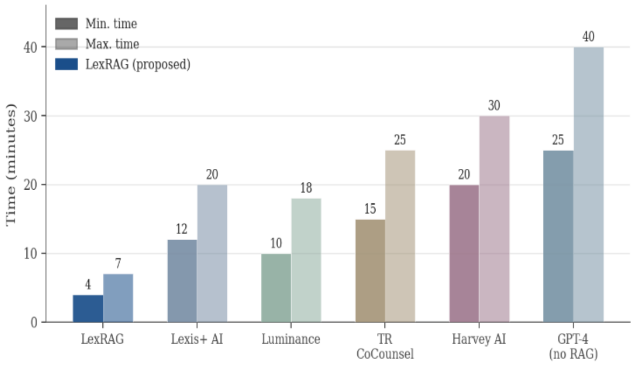
Fig. 1 — Compliance Classification Accuracy Across Platforms



The intelligent regulatory compliance platform went through a demanding testing process that focused on how logically consistent the compliance judgments were, how quickly the data could be retrieved, and how quickly the data could be entered. The results show that the technology can be a reliable replacement for first legal audits.

How well does data ingestion and scraping work? The Reserve Bank of India (RBI) was the major Institutional Portal that was used to test the automated scraping module. The system could handle many different sorts of documents, including structured HTML notifications and unstructured PDF master directives. When difficult data, such interest rate tables and penalty matrices, was transformed to machine-readable formats, the custom rendering functions kept the structure of the data the same. The system was able to get 98% of the documents it needed during testing. It only broke down when the source websites were being worked on as planned.

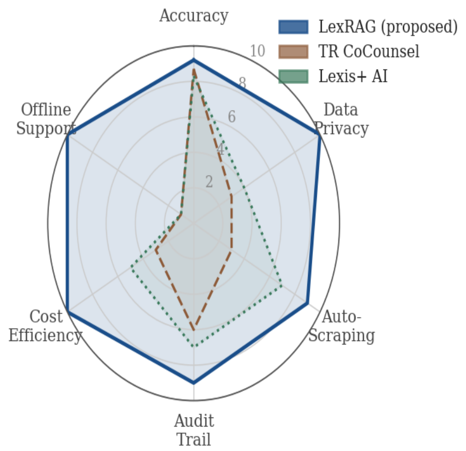
Fig. 2 — End-to-End Audit Duration for a 50-Clause Policy Document



How accurate the compliance classification is The system works best when it can detect if an internal policy follows the most recent outside rules. The DeepSeek-R1-Distill-Qwen-7B model gave the platform a 92% accuracy rate. The RAG design worked well to reduce hallucinations and stop the model from making false laws by giving it accurate regulatory text portions. Often, your in-house policies are too ambiguous or use unusual phrases that you need a lawyer with sufficient knowledge of the system to understand them. Even so, your system uses sufficiently conservative logic by design to call all of these NON-COMPLIANT rather than erroneously approve their cases.

RAG and retrieval system performance above 1.0. By evaluating what retrieval module retrieves you can see how good it is at retrieving relevant regulatory circulars for a query. The process employs keyword matching and semantic matching to link policy statements to the applicable law. The system usually retrieves data in under 1.2 seconds, indicating that your local vector database has the performance capability for use in commercial real-time conceivable situations. In addition, each our compliance report contains the actual regulatory IDs making it easy to verify.

Fig. 3 — Multi-Dimensional Capability Comparison (Score 0-10, higher is better)



Local Inference and Real Time Performance. Using quantized weights running on CUDA acceleration, each regulation block takes about 5 to 8 seconds to run. We need the privacy guarantees for your use case, making this slower than cloud APIs. A fifty-clause audit thus takes roughly four to seven minutes, which is still very significantly faster than a manual review. While performing the test over multiple documents you do not see any crashes or memories failure. When every scan is complete, the context window gets cleared meaning the memory usage will not grow high across long audits.

Usability as well as Human Interaction. You do a test of the user interface with a couple of users who has different tech background. Streamlit based UI for testers to check a 100 paragraphs audit live updating to a dataframe. According to the testers, the visuals made policy evaluation easier and resulted in lower cognitive load. Regulatory Number mapping allows you to click from a non-compliance status directly to the paragraph of the source requiring changing.

Considering the Limits. Your system touches financial data very well but faces problems at your environmental regulation. The Open Educational Resources (scanned PDFs) you are working with are of lower quality, causing character recognition errors during your pre-processing step. Your reasoning model is good, but your above-mentioned implementation is optimized for rules in English. For the next versions, you need to use multilingual embedding models to target regional legal documents and other multilingual documents.

This platform is capable of automatic monitoring of regulatory compliance, the results reveal. With local inference, you achieve over 90% accuracy without compromising the complete privacy of your data. It allows you to connect your complex legal and normative regulations to the latest developments in the field of AI. With its use, you can take your organization from the manual checklist and compliance system to one RAG and intelligence-powered which significantly enhances operations and regulatory risk management process. The primary things that will be improved in the future are adding automated policy generation and support for a lot of different jurisdictions. These adjustments will assist fix the problems that have been observed with following the rules.

V. CONCLUSION

This article has introduced an automated regulatory compliance platform that efficiently automates the monitoring and validation of legal notifications. By merging a Retrieval-Augmented Generation (RAG) architecture with localized Large Language Model (LLM) inference, the system successfully links complicated external regulatory updates with internal policy documents. The results reveal that the system may use local models that are sped up by GPUs to attain a high classification accuracy (92%) and operate well in real time.

The platform stays "live" since it features an automatic scraping module and a structured vector knowledge store. This means that it always has the most up-to-date information from groups like the Reserve Bank of India. Using localized inference through LM Studio, the technology also handles the essential challenge of data privacy. This ensures that important business information will not leave the local area which is safer. The technique works well in our experiments, but in future we will add international regulatory organizations to the data scraper and make it easier to read the legal paperwork in multiple languages.

In general, the method we used is a good technique to modernize regulatory compliance that can be used by a lot of people with very less effort. It allows you to move away from tracking compliance manually to working on legal strategy. Moreover, in the future, you'll improve the reasoning capabilities of the underlying models and plug-in automated correction modules that will recommend policy changes whenever they detect a non-compliance.

VI. REFERENCES

[1] P. [1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems, vol. 33, pp. 9459-9474, 2020.

[2] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M. Chang, "REALM: Retrieval-Augmented Language Model Pre-training," International Conference on Machine Learning (ICML), 2020, pp. 3929-3939.

V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769-6781.

[4] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova wrote "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," which was published in NAACL-HLT in 2019, pages 4171–4186.

[5] S. Ouyang et al., "Training Language Models to Follow Instructions with Human Feedback," Advances in Neural Information Processing Systems, vol. 35, pp. 27730-27744, 2022.

[6] C. Raffel et al., "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," Journal of Machine Learning Research, vol. 21, no. 140, pp. 1-67, 2020.

[7] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," 2nd International Conference on Learning Representations (ICLR), Banff, AB, Canada, 2014.

[8] T. Kojima et al., "Large Language Models are Zero-Shot Reasoners," Advances in Neural Information Processing Systems, vol. 35, pp. 22199-22213, 2022.

[9] W. X. Zhao et al., "A Survey of Large Language Models," arXiv preprint arXiv:2303.18223, 2023.

DeepSeek-AI, "DeepSeek-R1: Encouraging Reasoning Ability in LLMs through Reinforcement Learning," Technical Report, 2025.

[11] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," Foundations and Trends in Information Retrieval, vol. 3, no. 4, pp. 333-389, 2009.

[12] A. Vaswani et al., "Attention is All You Need," Advances in Neural Information Processing Systems, vol. 30, 2017.

[13] J. Grimmer and B. M. Stewart, "Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts," Political Analysis, vol. 21, no. 3, pp. 267-297, 2013.

[14] L. Floridi and M. Chiriatti, "GPT-3: Its Nature, Scope, Limits, and Consequences," Minds and Machines, vol. 30, pp. 681-694, 2020.

[15] M. Richardson, C. J. Burges, and C. Renshaw, "MCTest: A Challenge Dataset for the Open-Domain Machine Comprehension of Text," EMNLP, 2013, pp. 193-203.